

Sachin Bala

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Technical Skills

SolidWorks(3D), AutoCAD, Minitab, JMP, Python programming (Pandas, NumPy), SQL, LabVIEW, Six Sigma, Process Capability and Improvement, PFMEA, Lean Manufacturing, DOE, Microsoft Suites, Cognex, Dr Schenk vision system, Matlab,

Education

Western Michigan University (WMU)

Overall GPA: 3.69

Bachelor's in mechanical engineering with minor in Computer Science and Mathematics

Work Experiences

Associate Process Development Engineer, Tesla Inc., Austin, TX

02/25-present

- Commissioned two electrode lines producing electrode film at 75 meters per minute, achieving a 98% yield rate through systematic process optimization and rigorous qualification protocols.
- Implement lean manufacturing and Six Sigma methodologies (5S, FMEA, RCA, SPC, DOE, 8D, 5-Why, Capability Study) to enhance yield, increase manufacturing rate, and reduce downtime, significantly boosting operational efficiency.
- Develop robust and clear data collection, visualization, and analysis tools. Enable data driven operational and financial decisions through predictive insights into tool and process performance, including integration of factory data systems and use of software such as MySQL, Python, R, JMP, and Python.

Engineering Technician, Process Development Team, Tesla Inc., Austin, TX

8/23-02/25

- Created and deployed a pipeline in Grafana to monitor real-time tool vibrations, which improved process and tool analysis. Authored comprehensive SOPs for sensor installation and software usage, ensuring consistency and reliability.
- Qualified tools to meet stringent product specifications and process capabilities, achieving a 10% increase in tool production rate while maintaining high quality and overall cell performance.
- Conducted thorough root cause analysis of downtime issues and implemented corrective actions to minimize unplanned stoppages, leading to sustained improvements in tool reliability and operational uptime.

Engineering Co-Op, Scherdel SST, Muskegon, MI

8/22-12/22

- Researched EV components, tracked market trends and analyzed manufacturing materials for successful product launches, ensuring high-quality production processes.

Process Engineering Intern, Scherdel SST, Muskegon, MI

05/22-08/22

- Analyzed and optimized spring manufacturing process to catch early failure and built a predictive model to enhance production efficiency and accuracy by 25% by data analysis and ML algorithms.
- Programmed a smart Computer Vision system to catch the early failure in the product. Testing and Validation was done using Gauge R and R and F-1score analysis.

Laser Tech Technician Internship, Technique Inc., Jackson, MI

09/21- 04/22

- Optimized laser cutting using CAD and troubleshooting the machines to manufacture the vehicle chassis parts and validated them by design review and by using Laser tracker & polyworks.
- Updated SOPs and trained the technician and operators to follow standards.

Engineering Research and Project

Senior Design Project - Two Tier Industrial Cart (Technique Inc.)

01/22-12/22

- Designed safe and efficient two-tier industrial cart with Solidworks, integrating a pump system and height-adjustable scissor mechanism. Validated with FEA analysis and numerical calculations.
- DFMEA was implemented to enhance design efficiency by mitigating potential risk and adding safety features and a control plan.

Honors and Membership

Diether H. Haenicke Scholarship, WMU Dean's List,, Yellow Belt Training(Six sigma), Root Nepal (Co-Founder)